## 1. Project Objective

FinPulse is a full-stack stock market intelligence platform developed to monitor 20 NSE-listed companies, store historical market data ,expose REST APIs using FastAPI and visualize market information through an interactive React dashboard.

## 2. System Architecture

The application follows a three-tier architecture:

• Frontend: React + Vite for the user interface.
• Backend: FastAPI providing REST APIs and business logic.
• Database: SQLite for storing company fundamentals and historical OHLCV data.
• External Data Source: Yahoo Finance (yFinance).

## 3. Features Implemented

| Requirement | Implementation |
|---|---|
| Track 20 companies | Tracks 20 NSE-listed companies |
| Database | SQLite stores fundamentals and historical prices |
| REST APIs | 7 API endpoints for retrieval and updates |
| Historical Charts | Interactive line and candlestick charts |
| Fundamentals | Price, Market Cap, P/E Ratio, EPS |
| Comparison | Compare multiple companies |
| Dashboard | Interactive React dashboard with screener and market summary |

## 4. Database Design

Table: stocks

• ticker • company_name • price • market_cap • pe_ratio • eps • return_percent

Table: price_history

• ticker • date • open • high • low • close • volume

## 5. REST API Endpoints

GET /stocks
GET /stocks/{ticker}
GET /stocks/{ticker}/history
GET /market-summary
GET /compare?tickers=A,B,C

POST /update/{ticker}
POST /update-all

## 6. Challenges

The primary challenges included integrating React with FastAPI, handling incomplete responses from Yahoo Finance, designing an efficient schema for historical data, and rendering interactive financial charts while keeping the application responsive.

## 7. Future Improvements

Potential future enhancements include portfolio tracking, sector-wise analytics, real-time streaming market updates, user authentication, and downloadable reports.

## 8. Deployment

Frontend: Hosted on Vercel https://fin-pulse-n6o23jz27-sofi-844b.vercel.app Backend: Hosted on Render https://finpulse-backend-lghv.onrender.com The frontend communicates with the backend through REST APIs. The backend retrieves market data using Yahoo Finance (yFinance) and stores company fundamentals and historical price data in a SQLite database.

## 9. Conclusion

FinPulse successfully satisfies the assignment objectives by integrating data acquisition, persistent storage, REST APIs, and interactive visualization into a complete full-stack application. The project demonstrates practical experience with modern web development technologies and financial data processing.


Prepared By:
Sanchit Mishra
2025B5PS1392G